

ConvexOptimization 1

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PROBLEM I A set C is midpoint convex if whenever $a, b \in C$, the average or midpoint $\frac{a+b}{2} \in C$. Obviously a convex set is midpoint convex. It can be proved that under mild conditions midpoint convexity implies convexity. As a simple case, prove that if C is closed and midpoint convex, then C is convex.

SOLUTION. Need to prove $\forall a, b \in C, \forall \lambda \in (0, 1), \lambda a + (1 - \lambda)b \in C$. For convinence, write $f(\lambda) = \lambda a + (1 - \lambda)b$.

First, we prove $f(\lambda) \in C$ for $\lambda = \frac{k}{2^n}$ for $k = 0, 1, 2, \dots, 2^n$.

We prove it by induction. For $n = 1$, we have $k = 0, 1, 2$. For $k = 0, 2$, obvious. For $k = 1$, it's the definition of midpoint convex.

Assume for some $n = t \geq 1$, we've proved $f(\frac{k}{2^t}) \in C$, where $k = 0, \dots, 2^t$. Now consider $\lambda = \frac{k}{2^{t+1}}, k = 0, \dots, 2^{t+1}$. If $2 \mid k$, then $k = 2m$, then $\lambda = \frac{m}{2^t}, 0 \leq m \leq 2^t$. So by induction assumption $f(\lambda) \in C$. If $2 \nmid k$, then $2 \mid k - 1, k + 1$, and $0 \leq k - 1 < k + 1 \leq 2^{t+1}$. By induction assumption, we have $f(\frac{k-1}{2^{t+1}}) \in C, f(\frac{k+1}{2^{t+1}}) \in C$. Then $f(\frac{k}{2^{t+1}}) = \frac{1}{2}f(\frac{k-1}{2^{t+1}}) + \frac{1}{2}f(\frac{k+1}{2^{t+1}})$. By midpoint convex we have $f(\frac{k}{2^{t+1}}) \in C$.

By induction we have $f(\lambda) \in C$ for $\lambda = \frac{k}{2^n}, n \geq 1, 0 \leq k \leq 2^n$.

Since C is closed, for any $\lambda \in (0, 1)$, we can define $\lambda_n = \frac{\lfloor \lambda 2^n \rfloor}{2^n} \rightarrow \lambda$, and $f(\lambda_n) \in C$, and obviously f is continuous. So $f(\lambda) \in C$. So C is convex. $\square$

PROBLEM II $a, b \in \mathbb{R}^n, a \neq b$. Show $A := \{x \mid \|x - a\|_2 \leq \|x - b\|_2\}$ is a halfspace. Describe it explicitly as an inequality of the form $c^T x \leq d$. Draw a picture.

SOLUTION. $\|x - a\|^2 = (x - a)^T(x - a) \leq (x - b)^T(x - b) = \|x - b\|^2$, then rewrite the middle part, we get $(2b - 2a)^T x \leq \|b\|^2 - \|a\|^2$. $\square$

PROBLEM III Which of the following sets S are polyhedra? If possible, express S in the form $S = \{x \mid Ax \preceq b, Fx = g\}$.

1. $S = \{y_1 a_1 + y_2 a_2 \mid -1 \leq y_1 \leq 1, -1 \leq y_2 \leq 1\}$, where $a_1, a_2 \in \mathbb{R}^n$.
2. $S = \{x \in \mathbb{R}^n \mid x \succeq 0, 1^T x = 1, \sum_{i=1}^n x_i a_i = b_1, \sum_{i=1}^n x_i a_i^2 = b_2\}$, where $a_i, b_j \in \mathbb{R}, 1 \leq i \leq n, j = 1, 2$.

3. $S = \{x \in \mathbb{R}^n \mid x \succeq, x^T y \leq 1, \forall y : \|y\|_2 = 1\}$.
4. $S = \{x \in \mathbb{R}^n \mid x \succeq, x^T y \leq 1 \forall y : \sum_{i=1}^n |y_i| = 1\}$.

SOLUTION.

□

PROBLEM IV Finish the proof of the separating hyperlane theorem: Show that a separating hyperlane exists for C, D , where C, D are convex and $C \cap D = \emptyset$.

SOLUTION. Consider $S = \{x - y : x \in C, y \in D\}$, which is convex due to $\lambda(x_1 - y_1) + (1 - \lambda)(x_2 - y_2) = (\lambda x_1 + (1 - \lambda)x_2) - (\lambda y_1 + (1 - \lambda)y_2) \in S$. We only need to prove $\exists a$ satisfies $\{a^T z \leq 0 \mid z \in S\}$. Since $\forall z \in S$, if $a^T z \leq 0$, then $\forall x \in C, y \in D, a^T(x - y) \leq 0$, so $a^T x \leq a^T y$. So $\sup_{x \in C} a^T x \leq \inf_{y \in D} a^T y$. Let u satisfy $\sup_{x \in C} a^T x \leq u \leq \inf_{y \in D} a^T y$. Thus, $a^T x \leq u, \forall x \in C, a^T y \geq u, \forall y \in D$.

Let $d = \inf_{s \in S} \|s\|$. If $d > 0$, then $d = \inf_{s \in S} \|s\| = \inf_{x \in C, y \in D} \|x - y\| > 0$, we can find separating hyperlane to separate S and $\{0\}$, which is solved by 2.5.1 in textbook. So we only need to consider $d = 0$. Since $d = \inf_{s \in S} \|s\|$, we can get $0 \in \bar{S}$, where $\bar{S}$ is the closure of S . Another thing $C \cap D = \emptyset$, then $0 \notin S$, thus $0 \in S^c$. Since $0 \in \partial S$, where ∂S is the boundary of S , we can get $0 \in \partial S^c$. So $\exists t_k \in \partial S^c, k \in \mathbb{N}$ satisfies $t_k \rightarrow 0, k \rightarrow \infty$. Let $p_k = \arg \min_{p \in \bar{S}} \|t_k - p\|, k \in \mathbb{N}$ and p_k is well-defined due to $\bar{S} \cap \overline{B(0, \|t_k\|)}$ is compact, where $B(0, r) = \{x \mid \|x\| < r\}$. So we can get $0 \leq \|t_k - p_k\| \leq \|t_k\| \rightarrow 0$, then $0 \leq \|p_k\| \leq \|t_k\| + \|p_k - t_k\| \rightarrow 0$, that is $p_k \rightarrow 0$. Let $a_k = t_k - p_k, b_k = \frac{a_k}{\|a_k\|}, k \in \mathbb{N}$, then $\{b_k \mid k \in \mathbb{N}\}$ is a subset of unit ball. Due to unit ball is compact, unit ball is sequentially compact, thus $\{b_k \mid k \in \mathbb{N}\}$ contains a convergent sub-sequence $\{b_{k_i}\}_{i \in \mathbb{N}}$, whose limit is contained in unit ball. Let $b_{k_i} \rightarrow b$, where $\|b\| = 1$.

From here, we will prove $b^T z \leq 0, \forall z \in \bar{S}$. $\forall z \in \bar{S}, k \in \mathbb{N}$, since $p_k + \lambda(z - p_k) \in \bar{S}, \forall \lambda \in [0, 1]$ by convexity of S and the definition of p_k , we can get $\phi(\lambda) = \|t_k - (p_k + \lambda(z - p_k))\|^2 \geq \phi(0) = \|t_k - p_k\|^2, \forall \lambda \in [0, 1]$. Thus, we can get $\phi'_+(0) = \lim_{\lambda \rightarrow 0^+} \frac{\phi(\lambda) - \phi(0)}{\lambda} \geq 0$. Since $\phi'(\lambda) = -2(t_k - p_k - \lambda(z - p_k))^T(z - p_k)$, then $\phi'(\lambda)$ is continuous. So $\phi'(0) = \phi'_+(0) \geq 0$. That is $-2a_k^T(z - p_k) \geq 0$, so $a_k^T(z - p_k) \leq 0$, then $b_k^T(z - p_k) \leq 0$. So $b_k^T z \leq b_k^T p_k \leq \|b_k\| \|p_k\| = \|p_k\|$. Then $b_{k_j}^T z \leq \|p_{k_j}\|$. Since $b_{k_j}^T z \rightarrow b^T z, \|p_{k_j}\| \rightarrow 0$, we can get $b^T z \leq 0$. Therefore $\{b^T z \leq 0 \mid z \in S\}$.

□

PROBLEM V Let K^* be the dual cone of a convex cone K . Prove the following.

1. K^* is a convex cone.
2. $K_1 \subset K_2$ implies $K_2^* \subset K_1^*$.
3. K^* is closed.
4. $\text{int} K^* = \{y \mid y^T x > 0, \forall x \in K\}$.
5. If $\text{int} K \neq \emptyset$, then K^* is pointed.
6. K^{**} is the closure of K , i.e. $K^{**} = \bar{K}$.
7. If the closure of K is pointed then K^* has nonempty interior.

PROBLEM VI Define monotone nonnegative cone as

$$K_{m+} = \{x \in \mathbb{R}^n \mid x_1 \geq x_2 \geq \cdots \geq x_n \geq 0\}.$$

1. Show K_{m+} is a proper cone.
2. Find K_{m+}^* .